Sarathy Ranganathan

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Summary

Results-driven Digital Engineering with 7+ years of software development experience, delivering scalable and secure cloud-native solutions. Specialized in Python, and AWS, with a proven ability to lead and mentor engineering teams, drive end-to-end system design, conduct design reviews, and collaborate with cross-functional stakeholders to ensure timely delivery of high-quality solutions in the life sciences domain.

Skills

Languages & Programming : Python
Cloud & Infrastructure : AWS (Lambda, DMS, Glue, S3, Step Functions, SQS, SNS, RDS), CI/CD (GitHub Actions, Jenkins, Docker)
Databases : PostgreSQL, MySQL
Frameworks & Libraries : Pandas, Numpy, Django
Methodologies : Agile (Scrum), Test-Driven Development, Code Review & Quality Standards
Interpersonal : Technical Leadership, Stakeholder Management, Cross-functional Collaboration, Mentorship, Communication

Experience

Data Engineering, Management & Governance Senior Analyst
Accenture **Aug 2021 - Present, Chennai, IN**

- **Led a team** of 4 engineers on a **GxP-critical** Life Sciences project, owning SDLC documentation including technical design documents, test plans, and ensuring all artifacts met regulatory and product deployment standards.
- Architected **end-to-end architecture** and development of serverless, **cloud-native applications on AWS** (Lambda, EFS, S3) for Life Sciences clients, driving design decisions around scalability, security, and performance in production environments.
- **Designed** and optimized high-volume **data ingestion pipelines** using Python, ensuring accurate and timely data delivery aligned with business requirements.
- **Partnered** with **business stakeholders** to gather, clarify, and translate requirements into technical solutions, acting as the **primary bridge** between **functional teams and engineering** to ensure alignment throughout the project lifecycle.
- **Led CI/CD modernization** by migrating from Jenkins to GitHub Actions with OIDC-based authentication, reducing frontend **deployment time by 70%** and establishing best practices for automated, secure delivery pipelines.
- Conducted design reviews, enforced coding standards and testing strategies, and collaborated with cross-functional stakeholders including product owners and data scientists to deliver scalable data integration solutions.

Programmer Analyst
Cognizant **May 2019 - Aug 2021, Chennai, IN**

- Architected an NLP-based internal R&D POC for biomedical text analysis, building custom information extraction and semantic classification pipelines using Python.
- Engineered scalable FastAPI backend services to serve ML inference pipelines with asynchronous request handling, optimizing API performance for high-throughput data processing workflows.

Projects

FDA Submission Compliance Analyzer – Accenture

- **Architected and developed** a serverless backend **on AWS** using Lambda and Step Functions to orchestrate **multi-stage compliance workflows**, enabling automated consistency checks across multiple heterogeneous source file types at scale.
- **Identified a scalability bottleneck** in the existing single-part upload mechanism and **independently architected** a multipart upload pipeline supporting files up to 5GB, leveraging Amazon EFS for persistent storage, **eliminating file size constraints** and future-proofing the application's storage layer.
- Engineered a **fully automated CI/CD pipeline** using GitHub Actions with event-driven workflow triggers, replacing manual deployment scripts and **reducing the deployment time by 70%** (from ~20 minutes to under 5 minutes).
- **Eliminated** AWS Lambda **cold start overhead** by refactoring 20+ redundant Lambda layers into a single optimized shared layer, **reducing dependency resolution time** and **achieving a 20–40% reduction** in cold start latency.
- **Spearheaded CI/CD pipeline migration** from Jenkins to GitHub Actions in alignment with the organization's new security policy, implementing **OIDC role-based authentication** to enforce least-privilege access and improve deployment security and auditability.

Clinical Trial Randomization Data Management– Accenture

- **Reverse-engineered** & catalogued 6 Oracle database schemas in collaboration with business stakeholders, executing data discovery and filtration across **100+ tables** to identify migration-eligible datasets conforming to the target application's data model requirements **within a 3-month timeline**.
- **Independently evaluated and proposed** AWS DMS as the optimal CDC-based migration strategy, orchestrating a seamless data migration from Oracle source to PostgreSQL target, resolving cross-database data type incompatibilities across **~75GB of data records.**
- **Engineered multi-stage** post-migration **transformation workflows** in PostgreSQL, authoring 15+ complex SQL transformation scripts to **restructure and normalize** migrated datasets in compliance with the target application's business logic and data integrity constraints.
- Developed backend systems using Python on AWS Lambda and Django, designing and deploying serverless, event-driven data processing workflows that powered core application features and enabled seamless consumption of transformed datasets.

Education

Bachelor of Technology, Electronics and Communications Engineering
SRM University (2019)

Certifications

AWS Certified Developer - Associate (DVA)
Amazon Web Services (AWS) - 2025

Achievements

- **Accenture - The Extra Mile Award (for FY25 Q4),** acknowledging exceptional performance in couple of **GxP projects.**
- **Accenture - North Star - Pinnacle Award FY24 ,** recognized for taking initiative, going above and beyond, and consistently delivering.